Where AIXI and its variants sit in the arithmetic hierarchy (Section 13.1)

Δ_4^0

- π_M^* for infinite-horizon AIXI (after the $m \to \infty$ limit)

Δ_3^0

- π_M^* (optimal AIXI policy, Cor. 13.1.6)
- BayesExp (Thm 13.1.10)
- optimal knowledge-/entropy-seeking policy (Thm 13.1.9)

strictly harder (one more limit / quantifier)

Δ_2^0 **(approximable)**

- $\xi_U = M \in \Sigma_1^0 \subset \Delta_2^0$ (Solomonoff mixture)
- π_M^ε (ε-optimal AIXI policy, Cor. 13.1.6)
- ε-optimal knowledge-/entropy-seeking policy (Thm 13.1.9)

Δ_1^0 **(computable)**

- nothing: no π_ξ^*, no ε-optimal π_ξ^* for any semimeasure ξ (Thm 13.1.7, 13.1.8)